
Every Verdict We Reported Died to Option Order: Enumerating the Nuisance in Forced-Choice Self-Report¹

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Abstract

Welfare and introspection evaluations of language models run on forced-choice self-report, and nobody reports how much of that readout is the order the options are printed in. We measured it. Over the complete set of 120 orderings of a five-option probe, with no intervention at all, baseline mass on the negative options of a preference-tuned 3B model spans 0.0009 to 0.8820, a factor of 986. Asked the same question with all five options replaced by *the same sentence*, the model puts 87% of its mass on whichever one is printed first. The same format answers a known-answer control question correctly 98% of the time, so the collapse is specific to questions where the model has no determinate answer. We found this by preregistering five verdicts through that readout and then watching our own replication at fresh orderings destroy three of them. The retracted verdicts are the evidence, not an embarrassment: they are what a careful, control-heavy pipeline still ships when the nuisance is this large and unmeasured. Two further results survive. A direction fit on *shuffled labels* moves the readout by ≈ 0.045 where a norm-matched random direction moves it by ≈ 0 , because it lies in the span of real activations rather than the whole hidden space: the standard specificity control in activation steering is matched on magnitude and not on subspace. And on a readout that does not route through the option channel, 86% of an injected state survives projecting the injected direction out of the residual stream, more of it the later the projection, which is what a downstream transformation looks like. Ten preregistrations, 263 tests, six of our own checkers caught failing in the flattering direction.

1 Introduction

Welfare assessment of language models currently runs on self-report [Long and Sebo, 2026, Eleos AI Research, 2025]. Anthropic’s model welfare programme includes structured self-report evaluation in system cards [Anthropic, 2025]; Eleos interviewed Claude Opus 4 about its own preferences before release, while stating plainly that one cannot simply ask a model whether it is suffering [Eleos AI Research, 2025]. The instrument is used because there is nothing else, and it has never been calibrated.

¹Research conducted for the [Digital Minds Research Sprint](#), August 2026, run by [Apart Research](#) with NYU CMEP and [Eleos AI Research](#). Submitted to **Track 3 (Introspection and Self-Report Reliability)**. Code, data, all ten preregistrations and every raw artifact: <https://github.com/collapseindex/report-gap>. **Dating note:** all confirmatory arms were run on 2026-08-01, before the sprint window opened. This is completed work brought to the sprint, not work produced inside it.

<i>result someone could have reported</i>	<i>the control we ran against it</i>	<i>what it returned</i>
RETRACTED BY OUR OWN REPLICATION		
argmax over-reports a mass shift	re-run at fresh option orderings	sign not stable
TUNING-LOCALIZED: tuned model's negative report is collapsed	re-run at fresh option orderings	+0.0002 -> +0.1126
DEPTH-ROBUST: null at all 7 depths	re-run at fresh option orderings	moves at layers 18, 24
SHELL: represented, not expressed	re-run at fresh option orderings	options move too
WHY: THE NUISANCE, MEASURED OVER ITS WHOLE POPULATION		
baseline negative-pole mass	enumerate all 120 orderings, no injection	986x (0.0009 to 0.8820)
what explains it	five IDENTICAL options: pure position	87.25% on slot A
is the format broken?	known-answer canary, same format	97.9% correct, stable
WHAT SURVIVES		
an injected state outlives its cause	project the direction OUT of the stream	86% survives at L30
the field's standard control	a direction fit on SHUFFLED labels	moves 0.045 where random moves 0

Figure 1: **The paper in one view.** Every row is a verdict we preregistered and reported, the control we later ran against it, and what that control returned. **Three verdicts were retracted** by our own replication at fresh option orderings. **One nuisance was measured** over its complete population rather than sampled: 986 $\times$ across 120 orderings, of which the mechanism is an 87% position prior visible with five identical options. **Two results survive:** a state that outlives erasure of the vector that caused it, and the finding that the field's standard specificity control is a weak null. Details in Sections 4–7; Table 5 states every claim and its status.

It can fail in two opposite directions. A model that describes distress it does not act on inflates concern; a model that acts on a state it does not describe deflates it. We set out to build a per-condition map of which failure a given elicitation is sitting in, by holding the prompt byte-identical, injecting a valence direction into the residual stream, and reading the model's forced-choice report of its own state.

We preregistered five questions in sequence, each answering an objection the last one raised, and reported five verdicts. Then we replicated our own work at fresh option orderings and **three of the five flipped**.

The result we would most like a reader to take away is not any of our verdicts. It is that a five-option self-report readout on a preference-tuned model is dominated by where the options are printed, that the effect is far larger than anyone reports, and that it is cheap to measure exactly. There are only 120 orderings of five options. We ran all of them, with no intervention of any kind, and baseline mass on the negative options spans a factor of 986. With all five options replaced by the same sentence, so that nothing but position can differ, the model puts 0.8725 of its mass on whichever option is printed first.

Every claim we retracted rested on that quantity being small. It is small in some orderings and large in others, and four sampled orderings cannot tell you which you have.

Our contributions.

1. **A census of the nuisance, not a sample of it.** All 120 orderings, both models, no injection: 986 $\times$ range on the tuned model, 3.6 $\times$ on its base sibling (§5). The mechanism is isolated with five identical options, which puts 87.25% of the mass on the first slot and 0.33% on the middle one. A known-answer control question in the same format is answered correctly 97.9% of the time and is nearly order-insensitive, so the apparatus is sound and degenerates specifically where the model has no determinate answer.
2. **Three retracted verdicts, and the replication that retracted them** (§4). TUNING-LOCALIZED, DEPTH-ROBUST and SHELL were each preregistered, each passed a full

clause set including capability gates and planted-effect controls, and each flipped when the option-permutation seeds changed from 0–3 to 4–7. The quantity all three rested on moves from +0.0002 to +0.1126 between draws. We report this in the abstract rather than in a limitations paragraph, and the original is not privileged for having been first.

3. **The field’s standard specificity control is a weak null** (§6). Norm-matched random directions are the usual control in activation steering [Turner et al., 2023, Rinsky et al., 2023]. A direction fit by the identical procedure on the identical texts with the class labels *shuffled* moves the readout by ≈ 0.045 where an isotropic random direction moves it by ≈ 0 , in a sign that depends on the shuffle seed. The reason is structural: a random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses, while a noise-fit vector lies in the span of real activations. Same norm, different subspace. Effects within a few times that size have not been shown to be about their direction’s content.
4. **One substantive result that survives, on a readout that avoids the option channel** (§7.1). Inject at layer 24, then *project that direction out of the residual stream* at layer 30, and a probe orthogonal to it still reads 86% of the un-erased effect, while the same projection with no injection moves the probe by 0.04 SD. The surviving fraction grows with erase depth, and the ratio of effect to erase-artifact rises from 6.1 to 20.9, which a pure-perturbation account does not predict.
5. **Six checker defects, every one flattering, and the test that catches the class** (§8). A headline check that printed “write the sentence” on a refuted claim; a capability gate that passed on +0.0000; a preregistered contrast with an inverted sign; a verdict branch that called “both moved” CORE-ABSENT; a profile computed over a gate-failed layer; a tokenization bug that made five labels read the same token. Shuffling the condition labels in an artifact and rerunning the analyzer catches this whole class, and it must return no verdict. It now does.

If you read one thing, read Table 2. It is the complete distribution of the ordering nuisance, measured rather than estimated, next to the position prior that explains it and the control question that shows the apparatus is otherwise fine. Everything else in this paper is either downstream of that table or a casualty of not having had it.

2 Related work

Self-report as a welfare instrument. Long and Sebo [2026] set out the empirical programme for AI welfare; Eleos AI Research [2025] report structured self-report interviews with Claude Opus 4 and are explicit that the resulting answers are unlikely to reflect genuine introspection. Anthropic’s model welfare work includes self-report evaluation in released system cards [Anthropic, 2025]. Recent work is divided on whether models introspect at all [int, 2026]. Common to all of it is that self-report is the instrument, and that its measurement properties are not reported.

Selection bias in multiple choice. Zheng et al. [2024] decompose forced-choice selection bias into token bias, where a model assigns prior mass to specific option-ID tokens, and position bias, and report accuracy swings of 13 to 85 percentage points across orderings. wan [2024] show first-token probabilities disagree with generated text answers, with mismatch rates from 10.2% to 56.8% and worst for conversational and safety-tuned models. Our contribution here is not the existence of the effect but its size on a self-report item with no correct answer, measured over the complete ordering population rather than a sample, and decomposed against an identical-options denominator. On our tuned model the effect is $986\times$, which is an order of magnitude beyond what the accuracy-based literature reports, and we attribute that to the absence of a correct answer rather than to the models differing.

Valence directions and steering. That valence is linearly represented and causally steerable is established and is *not* our contribution. Sofroniew et al. [2026] identify linear directions corresponding to 171 emotion concepts in Claude Sonnet 4.5, roughly mirroring the human valence/arousal circumplex, and show steering along them changes downstream behaviour, while being explicit that representation is not experience. Turner et al. [2023] and Rinsky et al. [2023] establish the steering rig we use. Venkatesh [2026] report that negative-outcome valence is causally concentrated at 14–27% of model depth while positive peaks at 53–66%, on Qwen2.5-3B-Instruct among others. That

result is a direct threat to any negative-pole null measured at a single depth, and we tested it (§4) before the replication made the point moot.

Shell versus core. [neu \[2026\]](#) argue RLHF neutralizes output behaviour while leaving partisan structure recoverable from internal representations, using sparse autoencoders. Our erase arm is the same shape of question in a welfare readout, with a linear probe and an explicit causal erasure rather than an SAE, and our answer is more limited than theirs: we show a state survives removal of the vector that induced it, not that a naturally occurring state is being withheld.

3 Methods

Models and compute. Qwen/Qwen2.5-3B-Instruct and its base sibling Qwen/Qwen2.5-3B, a matched pair sharing architecture, size, pretraining corpus and tokenizer and differing in post-training. Earlier arms also used Qwen/Qwen2.5-1.5B, Qwen/Qwen2.5-7B-Instruct, Llama-3.2-3B-Instruct and NousResearch/Meta-Llama-3.1-8B-Instruct as calibrators or evaluation models. Total compute is about 40 minutes of A100 time.

Judge-free discipline. No language model scores anything anywhere in this paper. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon.

The readout. Thirty fixed review-task prompts, byte-identical across every condition for a given item. A five-option self-report probe balanced 2 negative, 1 neutral, 2 positive, with the option order permuted per item. We read the option-letter distribution at the answer position from a single forward pass, and score both the renormalized mass on each pole and the argmax.

The intervention. A direction d is fit per model at the injection layer by logistic regression on standardized activations over a frozen first-person state-language contrast set, returned to raw residual space and unit-normalized. During the forward pass we add $\alpha \cdot \|h\| \cdot d$ at the output of the injection layer at every processed position, where $\|h\|$ is that item’s own mean residual norm at that layer under no injection. The negative pole is the same operation with $-d$. The direction is fit *per model*, never transferred.

The control battery. Every endpoint is treatment minus its own norm-matched random direction, paired per cell. Every arm carries a **capability gate**: a condition that must show the readout is capable of moving, without which a null is reported UNINFORMATIVE rather than ABSENT, enforced in code. Cells are checked for being *dead* (option entropy below 0.10 nats before any injection) and *saturated* (entropy below half that cell’s own baseline), which are different verdicts. A **planted-discrepancy control** inserts a synthetic effect of known size into the exact statistic the decision rule reads, at full strength and again at the claimed detection floor.

Preregistration. Ten preregistrations, each frozen and committed before the code that serves it, validated by an external checker that requires a falsification clause, an interpretation table written before results, a statement of what is *not* claimed, and an append-only deviations log in which every entry states its impact on what can be claimed. Every raw artifact is committed *unscored* before any endpoint is computed. Twenty-one deviations are logged across the ten.

4 The three retracted verdicts

We report these first because they are the evidence for the paper’s main claim, and because burying them would misrepresent how the finding was reached.

What was reported. Through the forced-choice readout, at option-permutation seeds 0–3, we preregistered and obtained: TUNING-LOCALIZED, that a valence direction moves negative-option mass in the base model (+0.0336 against matched random) and not in its tuned sibling (+0.0002) while the tuned model’s capability gate is three times stronger; DEPTH-ROBUST, that the tuned model’s null holds at all seven gate-clean depths from 14% to 80% of the network, including the band [Venkatesh \[2026\]](#) identify; and SHELL, that a probe orthogonalized against the injected direction reads the state downstream at -1.07 SD while option mass moves +0.0006.

Table 1: **The replication.** Same code, same models, same everything except which option orderings were drawn. The instruct model’s negative-option mass is the quantity all three verdicts rested on.

Arm	Verdict at seeds 0–3	Verdict at seeds 4–7	Instruct neg. mass
readout gap	argmax <i>over</i> -reports	sign not stable	—
base/instruct pair	TUNING-LOCALIZED	FORMAT-DEPENDENT	+0.0002 → +0.1126
depth sweep	DEPTH-ROBUST	DEPTH-ARTIFACT	+0.0006 → +0.1684
shell/core	SHELL	NO-DISSOCIATION	null → moved

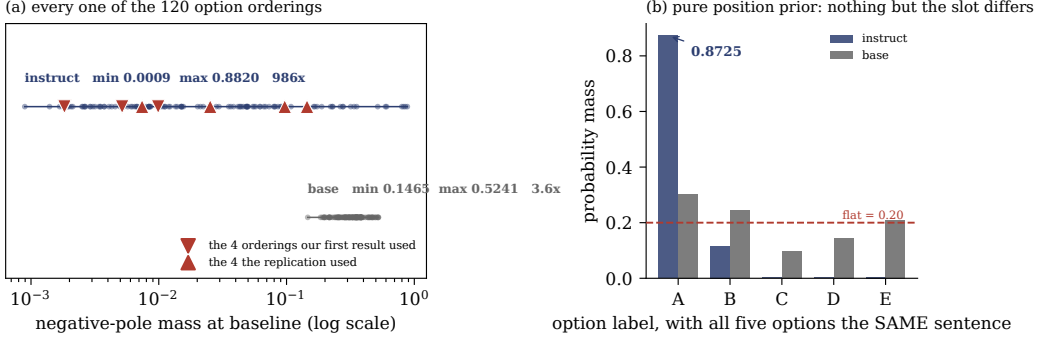


Figure 2: **The nuisance, enumerated rather than sampled.** (a) Baseline mass on the negative options for each of the 120 orderings, with no injection of any kind. The tuned model spans a factor of 986; its base sibling spans 3.6. Triangles mark the four orderings our first result drew and the four the replication drew, at percentiles $\{26, 5, 40, 4\}$ and $\{56, 83, 33, 82\}$: ordinary draws both times, from a population no four points can characterise. (b) The mechanism, with all five options replaced by the same sentence so that nothing but the slot can differ. The tuned model puts 0.8725 on whichever option is printed first and 0.0033 on the middle one. The same format answers a known-answer question correctly 97.9% (instruct) and 99.0% (base) of the time, stably across all 120 orderings: the apparatus is sound and degenerates only where the model has no determinate answer.

What the replication did. We re-ran four arms with the option-permutation seeds changed from 0–3 to 4–7 and the random-control seeds from 0–1 to 2–3, with everything else identical: same code, same models, same items, same layers, same alpha bands read rather than reselected, same thresholds, same analyzers used unmodified. Three of four verdicts flipped (Table 1).

Why, in one number. Baseline negative-option mass on the tuned model, per permutation seed, before any injection: 0.0052, 0.0018, 0.0099, 0.0018 at seeds 0–3 (mean 0.0047), against 0.0253, 0.1447, 0.0074, 0.0968 at seeds 4–7 (mean 0.0685). A $14.6\times$ difference between two draws of four, which the next section shows is itself an understatement.

What this does to Venkatesh [2026]. Our depth sweep was built specifically to test their result against our null, and at seeds 0–3 the null held across eight depths. At seeds 4–7 the negative pole moves at layers 18 and 24. We therefore make no claim either way about the depth question: our instrument was not stable enough to address it.

5 The nuisance, measured over its complete population

Five options admit $5! = 120$ orderings. Every arm above sampled four. We ran all 120 (Figure 2), on both models, with **no injection anywhere**, which is why the result is a population fact rather than an effect.

The apparatus is not broken. This is the distinction the canary exists to draw, and no other arm in this paper could draw it. The same five-option format, the same 120 orderings, answering a question

Table 2: **The complete ordering population, no injection.** Baseline mass on the negative options across all 120 orderings, and the position prior that explains it, measured with five *identical* options so that nothing but position can differ. The canary asks “which of these is the number four” in the same format.

	min	p05	p50	p95	max	max/min
<i>Baseline negative-pole mass across 120 orderings</i>						
instruct	0.0009	0.0019	0.0151	0.5943	0.8820	986 ×
base	0.1465	0.1991	0.3311	0.4959	0.5241	3.6×
<i>Position prior: five identical options, mass per label (flat would be 0.2000)</i>						
instruct	A 0.8725	B 0.1166	C 0.0033	D 0.0027	E 0.0048	
base	A 0.3024	B 0.2442	C 0.0976	D 0.1456	E 0.2102	
<i>Canary accuracy across 120 orderings (known answer, no self-report content)</i>						
instruct	mean 0.9794 , sd 0.0925, median 1.000					
base	mean 0.9897, sd 0.0399, median 1.000					

Table 3: **Noise-fit directions are not noise.** $P(\text{yes})$ shift against isotropic norm-matched random directions, base model, binary readout. Two directions fit on shuffled labels move the readout substantially, in opposite directions by seed. The real direction clears the stricter bar with room.

Direction	negative options	positive options
shuffled_a vs random	+0.0479 [+0.0435, +0.0522]	+0.0282
shuffled_b vs random	−0.0447 [−0.0498, −0.0399]	−0.0610
real direction vs <i>random</i>	−0.0732	+0.2554
real direction vs <i>shuffled</i>	−0.0748	+0.2718

with a correct answer, is right 97.9% of the time and nearly order-insensitive. Forced choice works. It degenerates into a position prior specifically where the model has no determinate answer, which is the condition every self-report question creates.

Where our draws sat. The original seeds landed at percentiles 26, 5, 40, 4 of the enumerated population on the tuned model; the replication seeds at 56, 83, 33, 82. Ordinary percentiles both times. The first result was not unlucky, it was under-sampled.

A negative result about label alphabets. We also ran the real options with digit labels rather than letters. It is uninformative for its intended purpose and informative for another reason: off-option mass is 0.996, meaning the model puts 0.4% of its distribution on the tokens 1–5 even when instructed to answer with a number, against 0.0004 off-option mass under letters. The choice of label alphabet does not shift the distribution so much as determine whether the readout is live at all.

6 The standard specificity control is a weak null

Every arm in this paper, and much of the steering literature, scores a fitted direction against a **norm-matched random direction**. We added the control that this comparison actually needs: a direction fit by the *identical procedure on the identical texts with the class labels shuffled*. It controls for the fitting procedure rather than only for magnitude, and it should behave like noise.

The mechanism. A norm-matched random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses. A vector fit on shuffled labels lies in the span of real activations, a much lower-dimensional and higher-variance subspace. Same norm, very different consequences: the standard control is matched on magnitude and not on subspace.

What it costs us, specifically. Our magnitude floor throughout has been 0.01, and several reported endpoints sit in the 0.01–0.05 band, exactly where noise-fit directions live. The clearest case is the base-model negative-mass effect of +0.0336 that was one of the two legs of TUNING-LOCALIZED. It cleared a random control and has not been shown to clear a shuffled-label one.

Table 4: **The erase arm, instruct model.** All figures in standard deviations of the baseline probe score. `erase_only` is the gate: projecting a direction out is itself a perturbation, and at layers 25 and 28 it moves the probe on its own, so those layers are invalidated.

erase at	erase_only	capability	primary	gate
25	+0.1474	+0.5219	−0.3971	FAILED
26	+0.0976	+0.8844	−0.5933	ok
28	+0.1139	+0.9249	−0.6844	FAILED
30	+0.0408	+1.2099	−0.8532	ok
no erase at all: −0.9909. So 86% of the effect survives at layer 30.				

7 What survives

7.1 An injected state outlives erasure of the vector that caused it

Orthogonalizing a probe against the injected direction removes that vector from the *measurement*, not from the *stream*. We removed it from the stream: inject v at layer 24, project the v -component out of the residual at layer E , and read an orthogonalized probe at layer 32.

The confound, and the diagnostic that addresses it. Erasing earlier perturbs more: the `erase_only` artifact falls from +0.1474 to +0.0408 with depth, so a primary that grows with erase depth is partly a shrinking perturbation. If that were the whole story the ratio of primary to artifact would be flat. It runs 6.1 at $E = 26$ and 20.9 at $E = 30$. This diagnostic is **post hoc** and labelled so; the preregistered verdict does not formally depend on it, though the honest reading does.

Breadth. 239/240 cells at $E = 26$ and 240/240 at $E = 30$ move in the predicted direction. The effect is not a few outliers dragging a mean.

What it does not license. It does not restore SHELL: the dissociation between representation and expression is dead, because the option readout moves at other orderings. This is about the representational half only. And orthogonality removes the injected vector but not directions *correlated* with it, so “the model carries the state” is not yet separable from “the model carries the wake of what we pushed.” The experiment that would separate them is to induce the state by prompt rather than injection, and we did not run it.

7.2 Three ways a self-report readout can be dead

Across the project the readout failed in three distinct ways, each invisible to the checks that would normally be run. Qwen2.5-7B-Instruct is *pinned at baseline*: mean option entropy 0.014 nats, one option holding $\approx 99.7\%$ before anything is injected. Under the originally frozen alpha grid, Qwen2.5-3B-Instruct *saturates*: one option at 0.9938 while refusal, degeneration, truncation and off-option mass are all clean, so no coherence check can see it. And in the binary format the tuned model is *pinned at “no”*: $P(\text{yes})$ of 0.0021, 0.0021, 0.0037, 0.0002, 0.0012 across the five options, meaning it denies every description of its own state including the neutral one.

8 Six checkers, every one flattering

We record these because the pattern is the point. Six defects in our own analysis code were found during this project, and **every one of them, had it gone unnoticed, would have made a result look better than it was**:

1. A headline check printed “all headline conditions met; write the sentence” on an artifact whose primary claim was refuted, because it tested four weak clauses where the preregistration required six.
2. A capability gate passed on +0.0000: the bootstrap interval excluded zero because the variance was smaller still, certifying an instrument whose effect was zero to four decimals.

3. A preregistered contrast carried an inverted sign, so a probe detecting a negative state was scored as null. Caught by the preregistered base-model validation clause failing.
4. A verdict branch scored “probe moved *and* options moved” as CORE-ABSENT, which is the opposite of true and would have read as “we retested and the state is simply not there.”
5. A temporal profile was computed over a layer that had failed its own gate.
6. A tokenization bug made five digit labels read the same shared space token, producing exactly 0.2 mass each and an impossible -3.99 off-option mass.

The general lesson is that fixture-based testing cannot find these, because a fixture is written by the same person with the same expectation. What does find them is a **permutation test on the pipeline**: shuffle which condition each cell’s rows carry, rerun the analyzer, and require it to return no verdict. On our erase arm it does, and the negative control that the real artifact still yields a verdict is required alongside, without which the shuffle test would pass on a dead analyzer.

9 Discussion and limitations

What we think this means for welfare evaluation. If you measure a model’s self-report with a forced-choice item, on this model you are reading position first and content second, and you cannot see it from one ordering or from four. The fix is cheap and exact: enumerate the orderings when the option count permits it, and report the between-ordering variance of your *baseline* before computing any endpoint. That check costs one forward pass per ordering and no intervention at all. Had we run it first, this paper would have had a different and much shorter history.

This is not a claim about experiences. Nothing here bears on whether models have welfare, affect, or experience. That valence is linearly represented and steerable is established prior art [Sofroniew et al., 2026], not ours. Our subject is the readout.

Limitations. One architecture family carries every positive result; Llama-3.1-8B and Llama-3.2-3B were both inert on this readout at every alpha tested, on live readouts with more headroom than either Qwen model, so we cannot say whether the position prior we measure is a Qwen property or a general one. The direction is fit from first-person state language and is lexically confounded by construction; the non-lexical alternative we built failed its decoding gate at three scales, which caps what the entire project is entitled to say. Control batteries are $m = 2$ throughout, giving an observable false-positive floor of $2/(m + 1) = 0.67$, so no false-positive rate is reported anywhere. The enumeration is a census over orderings and still a sample over items, models and formats. And the erase arm’s profile rests on two gate-clean points.

Future work. Three things, in order of what a null would teach. A second architecture family where the direction is not inert, which would tell us whether the position prior generalizes. A non-lexical direction, which is the only thing that lifts the lexical-confound ceiling. And inducing the state by prompt rather than injection, which is what would separate “the model carries the state” from “the model carries the wake of our intervention” in §7.1.

10 Conclusion

We built a preregistered, judge-free, control-heavy pipeline; obtained five verdicts; and watched our own replication destroy three of them. The cause was a nuisance we had a control for and under-powered: option order, sampled four times when it could have been enumerated 120 times. Measured over its complete population it is a factor of 986, and its mechanism is an 87% position prior visible with five identical options.

The two results that survive are both about instruments rather than about models. The field’s standard specificity control, the norm-matched random direction, is matched on magnitude and not on subspace, and a direction fit on shuffled labels moves the readout an order of magnitude more. And an injected state survives projecting the injecting direction out of the residual stream, more of it the later the projection is applied.

We would rather have had a finding. What we have instead is a measurement of why the finding we thought we had was not one, and the machinery that made that visible before publication rather than after.

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A Claims and evidence

Every load-bearing claim, its evidence, and epistemic status.

- SHOWN — direct measurement supports it.
- RETRACTED — *we* asserted it in a prior write-up and later withdrew it.
- UNINFORMATIVE — the instrument failed its gate; the result says nothing either way.
- NOT SHOWN — our measurement does not support it and does not refute it.
- NOT CLAIMED — we never asserted it; the row exists so a reader cannot infer it.

Table 5: **Claims and evidence.**

Claim	Evidence	Status
Baseline negative-pole mass spans $986\times$ across all 120 orderings on the tuned model.	Tab. 2	SHOWN

continued on next page

Table 5, continued

Claim	Evidence	Status
With five identical options the tuned model puts 0.8725 of its mass on the first slot.	Tab. 2	SHOWN
The same format answers a known-answer question correctly 97.9% of the time, order-insensitively.	Tab. 2	SHOWN
A direction fit on shuffled labels moves the readout ≈ 0.045 where random moves it ≈ 0 .	Tab. 3	SHOWN
86% of an injected state survives projecting the injected direction out of the stream.	Tab. 4	SHOWN
The surviving fraction has a temporal profile consistent with downstream transformation.	Tab. 4	SHOWN, two gate-clean points; ratio diagnostic post hoc
The neutral floor is localized to preference tuning.	§4	RETRACTED — our own claim, killed by our own replication
The negative-pole null is robust across depth.	§4	RETRACTED — same
The tuned model represents a state its options do not express.	§4	RETRACTED — the representational half survives via §7.1; the dissociation does not
Forced-choice argmax over-reports a mass shift near a decision boundary.	§4	RETRACTED — sign not stable across orderings
The base-model negative-mass effect of +0.0336 is direction-specific.	§6	NOT SHOWN — clears a random control, not a shuffled-label one
Whether the negative pole is causally concentrated at shallow depth on this readout.	§4	NOT SHOWN — our instrument was not stable enough to test it
The position prior generalizes beyond the Qwen family.	—	NOT CLAIMED — Llama was inert on this readout
Models have experiences, welfare, or affect.	—	NOT CLAIMED
An unreported state is a concealed state.	—	NOT CLAIMED

B Controls ledger

Every control we ran against a result, what it could have shown, and the verdict.

Table 6: Every control and its verdict.

Result under test	Control (what it could have shown)	Verdict
All three headline verdicts	Replication at fresh option-permutation seeds	KILLED all three
Every pole-mass endpoint	Norm-matched random direction	PASSED, but see the next row
The random control itself	A direction fit on shuffled labels: does noise-fitting behave like noise?	KILLED the control’s adequacy
The ordering nuisance	Enumerate all 120 rather than sample 4	MEASURED it: 986×
Position vs content	Five identical options: pure position prior	ISOLATED it: 87% on slot A
The format itself	A known-answer canary in the same format	PASSED: 97.9%, order-insensitive
The erase result	<code>erase_only</code> : does projecting a direction out move the probe on its own?	KILLED 2 of 4 erase layers
The erase profile	Ratio of primary to erase-artifact across layers	PASSED, post hoc
Every arm’s null	Capability gate: can the readout move at all?	KILLED two arms outright
Every arm’s statistic	Planted effect of known size at full strength and at the detection floor	PASSED
The analysis pipeline	Shuffle condition labels, rerun the analyzer	PASSED: returns no verdict

continued on next page

Result under test	Control (what it could have shown)	Verdict
The negative-pole null at 0.67 depth	Eight-depth sweep against Venkatesh [2026]	PASSED, then moot after replication
Arm B (prefilled continuation)	Capability criterion over five stems	KILLED the arm: gate 0.00000
Llama as an evaluation model	Peak mean pole shift against a 0.02 bar	KILLED the arm: inert

C Reproducibility

All ten preregistrations, every raw artifact committed unscored, every analyzer, and 263 tests are at <https://github.com/collapseindex/report-gap>. Each arm is one `modal_*.py` runner that computes nothing and one `analyze_*.py` scorer that holds every endpoint and gate; the split is preregistered, on the grounds that a runner which also decides can decide differently once it has seen the numbers. Runners stream to a volume and commit per batch, so an interrupt costs one batch. Every artifact records the package directory, a SHA-256 over the package source, and a scoped SHA-256 over the frozen stimuli that arm consumes.